

A systematic review of phosphogypsum recycling industry based on the survey data in China – applications, drivers, obstacles, and solutions

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ABSTRACT

The development of phosphogypsum (PG) recycling industry plays an important role in promoting the environmentally friendly transition of the phosphorus chemical industry, worldwide. China is the leading producer of phosphate and PG in the world, as the development of PG recycling industry in China has already generated very promising outcomes. Over 80% of China's total PG production is contributed collectively by four provinces, including Sichuan, Hubei, Guizhou, and Yunnan. By conducting an in-depth investigation to 18 representative companies involved in PG recycling allocated in these four provinces, the purpose of this study is acquired a comprehensive insight into the status quo of the development of PG recycling industry in China and its prevalent challenges. The results indicate that, first, recycled PG can be used in various applications, which predominantly encompassing in cement retarder, construction materials, engineering application, polymer materials, underground back filling, ecological restoration, sulfur acid, fertilizers, soil amendment, crafts, etc. Second, the key drivers behind the development of PG recycling industry include types of companies, technological innovation, products and markets, industrial policies, and geographical location. Third, the obstacles hindering the development of PG recycling industry include: a scarcity of innovative PG recycling companies; an absence of comprehensive technical evaluation standards and high-value technological breakthroughs; challenges in promoting PG-based products in the market; insufficient support from industrial policies; and the lack of geographical advantages contributing to sluggish industrial progress in specific regions. Fourth, the recommended corresponding solutions encompass: facilitating the transformation and upgrading of phosphorus chemical companies and promoting the coordinated development of PG recycling companies; instituting a technical evaluation standardized system and promoting high-value technologies; broadening the market channels and establishing a reproducible and scalable PG recycling industrial development model; accelerating the enhancement of industrial policies; and developing plans suitable for regional PG recycling industry based on local conditions.

1. Introduction

Phosphogypsum (PG) is the major by-product of the wet phosphoric acid industry (Li et al., 2023), which is primarily composed of calcium sulfate dihydrate, along with some impurities including soluble phosphorus, fluorine, and heavy metals (Ou et al., 2023). Approximately 4–5 tons of PG are generated from the production of phosphoric acid per ton (Yin et al., 2023). Roughly estimated, about 2.8×10^8 t of PG are produced globally each year, where only around 15% is recycled and

applied in various fields such as construction materials, agriculture, etc. (Silva et al., 2019). In countries such as the United States and Canada PG is mainly stockpiled (Hilton, 2020). Currently, the worldwide PG stockpiles has surged to approximately 40×10^8 t (Birky et al., 2016). This extensive accumulation of PG imposes significant impacts on ecology and environment, such as land occupation, soil pollution, groundwater contamination, resource depletion, ultimately endangering people's life and property (Nelson et al., 2021). In order to enhance the ecological efficiency of the phosphorus chemical industry,

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it's crucial to promote the advancement of PG recycling industry. According to the United Nations Sustainable Development Goals (UNSDGs), PG can be recognized as a valuable asset within the phosphorus chemical industry, rather than industrial solid waste. Recycled PG can be reutilized in many fields to make great contribution on global secondary resource inventory for sustainable development (Hilton, 2020).

Phosphorus chemical companies in China are primarily clustered within phosphorus-rich provinces like Sichuan, Hubei, Guizhou, and Yunnan, with an annual production of 75×10^6 t of PG in 2020 (Cui et al., 2022b). Under the combined pressures of ecological environmental preservation and rising economic costs, the structure of PG recycling industry in China was established with a wide coverage. According to the report released by the International Fertilizer Association (IFA) in 2020, the annual PG recycling rate can reach approximately 40%. (Hilton, 2020). Despite positive achievements, a portion of 60% in annual PG production is unutilized, still. Hence, considerable endeavors are essential to drive the industrialization of PG recycling industry. In 2021, in order to further expand the applications of recycled PG, a directive, the *Guiding Opinions on Comprehensive Utilization of Bulk Solid Waste for the 14th Five-Year Plan*, was released by the National Development and Reform Commission of China (National Development and Reform Commission, 2021). Particularly, it was emphasized to explore the applications of recycled PG in specific fields, such as construction materials, soil amendment, underground backfilling, and roadbed materials.

Currently, researches on PG recycling are primarily focusing on in-depth exploration of various application technologies for recycled PG and their environmental impacts (Attallah et al., 2019; Cao et al., 2021). For example, application technologies include PG pretreated with water washing as aggregate for cemented backfilling (Zhou et al., 2022), preparation of gypsum with high purity and whiteness from PG (Zhang and Fan, 2023), and utilization of PG in the production of white textured paints (Valančius et al., 2022). Regarding environmental impacts, Majdoubi et al. (2023) explore the potential and environmental assessment of acidic geopolymer technology in PG recycling. Chen et al. (2018) analyze the environmental impact of PG are recycled for cemented paste backfill technology.

However, the PG recycling involves not only technical and environmental impact concerns, but also the consideration of economic aspects. Liu et al. (2023b) focused on the cemented paste backfill technology and summarized the profitability and sustainability of PG in rare earth elements recovery, construction materials, and agricultural applications. To achieve the industrialization of PG recycling industry, a range of factors, including policy guidance and the coordinated development of the entire industry chain, need to be seriously considered (Cui et al., 2022a). Currently, there is a lack of studies that thoroughly analyze the development of PG recycling industry from a comprehensive standpoint. What is the current status of the PG recycling industry, what are the challenges it is confronting, and how can these challenges be addressed? Hence, through field inspection and questionnaire to 18 selected companies engaged in PG recycling allocated in four major provinces in China, this study systematically analyzes the applications, drivers, obstacles, and solutions of PG recycling industry in China. The goal is to offer insights that could serve as a reference for fostering the sustainable and robust growth of PG recycling industry in China and globally.

2. Literature review

2.1. The production and hazards of phosphogypsum

Based on the China Industrial Statistical Yearbook and China Phosphate and Compound Fertilizer Industry Association, Hou et al. (2022) calculated that China's annual production of PG reached 62×10^6 t in 2010 and increased to 75×10^6 t in 2020, an increase of 20.97%. The

yearly rise in PG production has given rise to a range of resource and environmental contamination issues (Melki and Gueddari, 2018). Du et al. (2020) summarized the composition of PG as comprising phosphorus impurities, fluorine contaminants, organic pollutants, and additional impurities. Excessive phosphorus impurities entering surface waters can lead to severe eutrophication at the watershed scale (Yuan et al., 2018; Liu et al., 2020). High fluoride intake can increase the risk of chronic toxicity and cancer in humans (Liu et al., 2022). Organic pollutants (flocculants, organic catalysts, etc.) can diminish the strength and gelling characteristics of PG, affecting the market acceptance of PG-based products (Deng et al., 2023). And impurities of heavy metals (Ba, Cu, Pb, Zn) and toxic radionuclides (Ra^{226} , U^{238} et al.) can affect the performance of PG (Bisone et al., 2017). The presence of various impurities severely limits the recycling of PG.

2.2. Phosphogypsum recycling

Due to the existence of toxic and harmful substances in PG, roughly 85% of the global PG is primarily stockpiled, with only 15% of it can be recycled (Silva et al., 2019). Currently, the PG recycling worldwide mainly find application in three key sectors: construction materials, chemical industry, and agriculture (Liu et al., 2023a; Bounaga et al., 2022). Calderón-Morales et al. (2021) found that PG-based cementitious systems can achieve high compressive strength. Weiksnar et al. (2023) proposed the utilization of OG as an amendment for graded aggregate road base to replace traditional road base materials. Microbiological processes can valorize PG, particularly by recovering sulfur in sulfate as elemental S, which can be used to make the sulfuric acid (Bounaga et al., 2022). Abril et al. (2009) analyzed PG as a Ca amendment in saline marsh soils of southern Spain. Research on PG recycling mostly focuses on the physical and chemical properties and technical principles of various PG applications. What is notably absent is a comprehensive review of specific applications and its advantages, which is the foundation for exploring the development of PG recycling industry.

2.3. The development of phosphogypsum recycling industry

The sustainable development of PG recycling industry is a significant approach to achieve circular economy and waste management of PG recycling (Wang et al., 2021). The Scottish Waste Environmental Footprint Tool proposed by Salemddeb et al. (2021) can help policymakers understand the potential environmental impacts of policy options. Cui et al. (2022b) assessed the effects of policy intervention on PG recycling industry through the construction of a dynamic business model. Through a review of PG samples from 67 industrial storage sites around the world, Bilal et al. (2023) analyzed the PG circular economy considerations. Current research typically considers the environmental, policy, or economic implications of PG recycling from a single aspect. However, the development of PG recycling industry encompasses numerous facets. Existing research has not comprehensively analyzed the factors influencing the development of PG recycling industry, along with the challenges and potential solutions encountered during its progress.

Enlightened by literature reviews, the innovation of this research is to systematically summarize the applications of PG recycling in China, as well as the drivers, obstacles, and solutions for the development of PG recycling industry through field inspection of companies related to PG recycling in China, which can contribute to PG recycling industry management in the future.

3. Methodology

The field research method is to apply an objective attitude and scientific methods to conduct on-site inspections within a defined scope to identify and address problems (Sanjek, 2001). This approach finds extensive application in various fields, including sociology and

management (Sanjek, 2015). The main distribution of PG recycling industry is concentrated in Sichuan, Hubei, Guizhou, and Yunnan Provinces, accounting for more than 80% (Cui et al., 2022b). This study uses field research method to investigate 18 companies related to PG recycling in these four provinces from 2020 to 2022. The purpose is to deeply understand the current situation of PG recycling industry in China, explore the drivers and obstacles for the development of PG recycling industry, and propose corresponding solutions. The data comes from surveys of these 18 representative companies (Appendix Table). These companies involve three types: PG production companies, PG production & recycling companies, and PG recycling companies. The annual production of PG by these 18 companies is 34.74×10^6 t, constituting 46.31% of the national total of 75×10^6 t in 2020 (Cui et al., 2022b). The annual recycling of PG by these 18 companies reached 19.26×10^6 t, achieving a recycling rate of 55.44%. The survey data of these companies can effectively reflect the current developmental status, prevalent challenges, and evolving trends within PG recycling industry in China.

To offer a more visual analysis of PG recycling industry in China, this study uses ArcGIS to create geographical maps showcasing the locations of these 18 representative companies. The amount of PG production and recycling for each company (Unit: 10000 t/y) and applications are indicated, as shown in Fig. 1.

As shown in Fig. 1, applications of PG recycling in China mainly include cement retarder, construction materials, engineering application, polymer materials, underground filling and ecological restoration, sulfur acid, fertilizers, soil amendment, crafts, etc. Among these, certain applications have undergone successful testing but are yet to be implemented in production.

Sichuan Province exhibits higher terrain in the western regions and

lower terrain in the east. PG-related companies are mainly located in relatively flat areas like Chengdu, Mianyang, Deyang City, characterized by intersecting river networks. The applications of PG recycling in Sichuan Province are mostly cement retarder and construction materials. Hubei Province is primarily characterized by plains and hills with relatively low altitudes. PG-related companies are concentrated in Yichang City, which has developed water transportation. The applications of PG recycling in Hubei Province are also cement retarder and construction materials, while engineering application is in the experimental phase. Guizhou Province is situated in the eastern region of the Yunnan-Guizhou Plateau, characterized by intricate terrain and Karst landform. PG-related companies are clustered in Guiyang City and its surrounding cities like Qiannan City. The applications of PG recycling in Guizhou Province encompass cement retarder, construction materials, engineering application, polymer materials, underground filling, sulfur acid, and fertilizers. Yunnan Province is characterized by a mountainous plateau landscape with steep terrain, high altitude, and abundant phosphate mineral resources. PG-related companies are situated around Dianchi Lake in Kunming City. The applications of PG recycling in Yunnan Province include cement retarder, construction materials, and engineering application, while polymer materials, ecological restoration, soil amendment, and crafts are in the experimental phase.

4. Results

4.1. Applications of phosphogypsum recycling

Fig. 2 shows the distribution of PG recycling for each application that have been implemented in surveyed region. Derived from the survey data, the annual production of PG in Sichuan Province is relatively

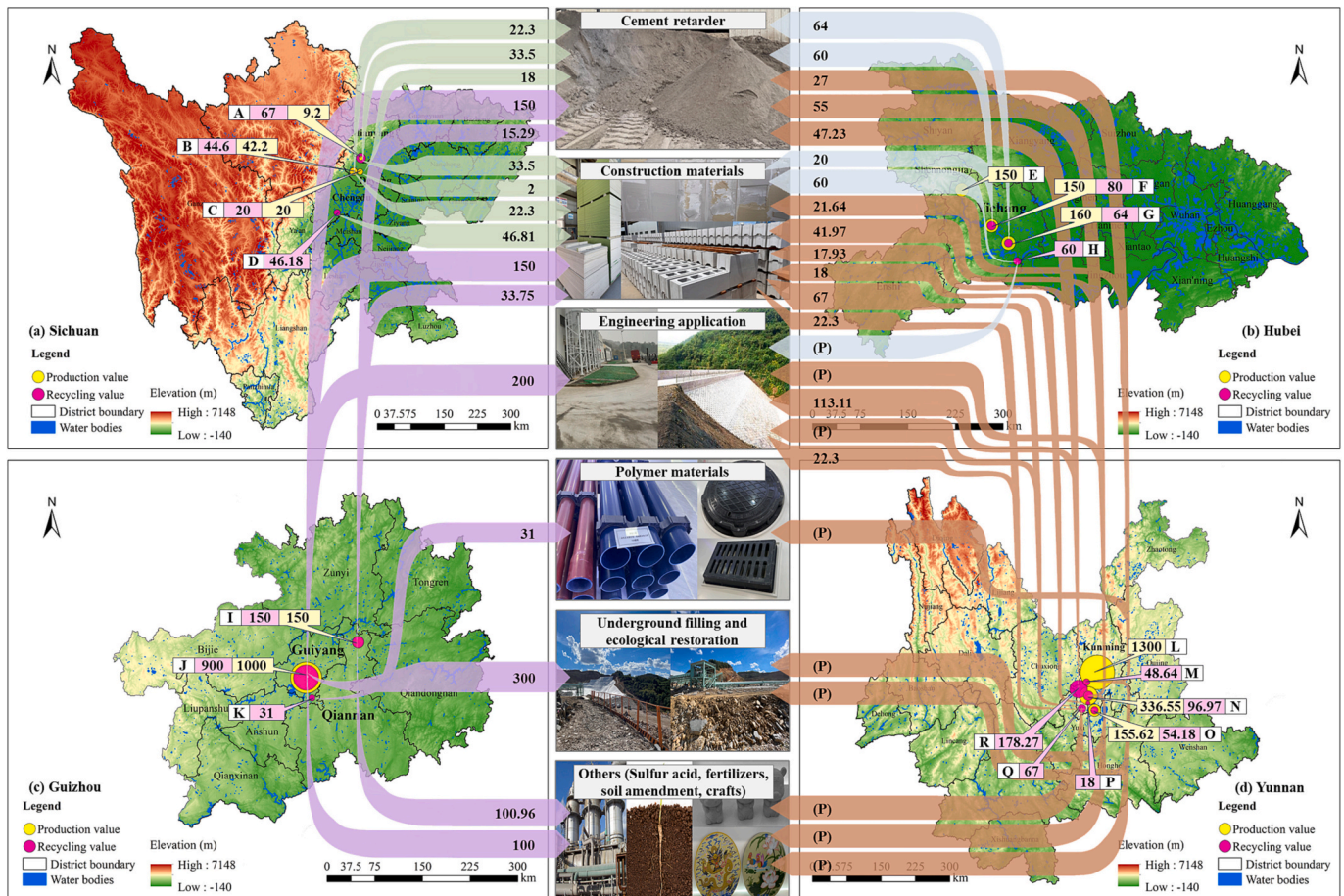


Fig. 1. Geographical distribution, PG production and recycling of surveyed companies.

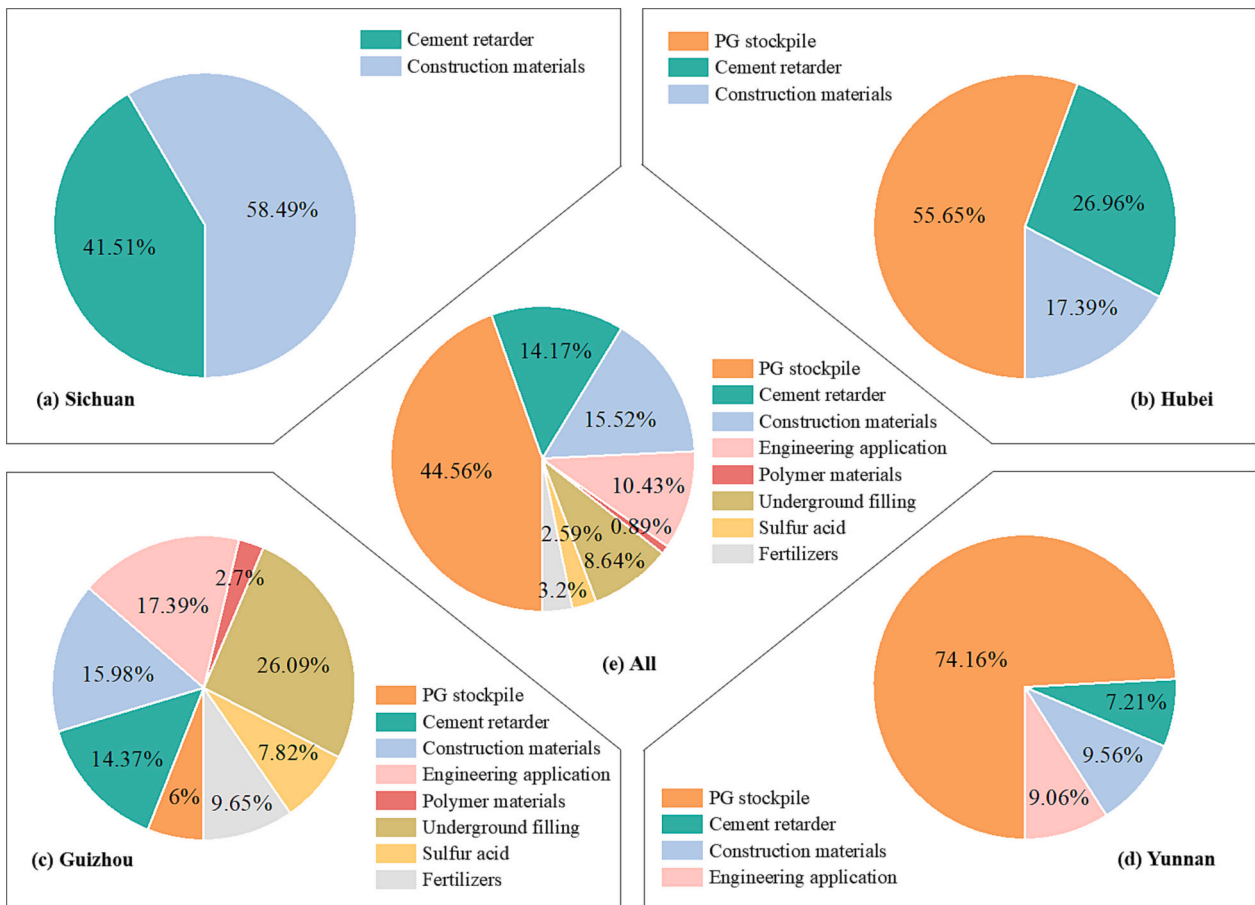


Fig. 2. The Pie chart for the distribution of PG recycling based on survey data.

small, it has successfully achieved 100% PG recycling. And the formerly stockpiled PG has commenced its consumption. In Hubei Province, the recycling rate of PG stands at 44.35%. A majority of PG-related companies are situated alongside the Yangtze River, capitalizing on the clear advantage of water transportation. Due to the low quality of PG, the applications still based on traditional cement retarder and construction materials. Supported by robust policy measures and economic incentives, Guizhou Province has successfully attained a recycling rate of 94%. It has pioneered the exploration of diverse new applications of PG recycling. Owing to the lack of convenient transportation, inadequate policy support, and substantial PG production in Yunnan Province, the recycling rate is merely 25.84%. Overall, the recycling rate of PG in the surveyed region is 55.44%, and applications of PG recycling are as follows.

(1) Cement retarder

Through processes including washing, flotation, ball milling, and calcination, PG undergoes pretreatment (Li et al., 2022; Palla et al., 2022). This treatment enables PG to effectively take the place of natural gypsum and desulfurized gypsum, functioning as a regulator for the setting time of cement (Qin et al., 2023). In the surveyed region, 14.17% of the PG is applied for cement retarders, as shown in Fig. 2, ranking second only to construction materials, which accounts for 15.52%. Because of its uncomplicated chemical modification methods, cost-effective production, and well-established production process, cement retarders remains the primary application for PG recycling in China (Li et al., 2022).

(2) Construction materials

PG takes the place of natural gypsum in the production of gypsum powder (Jin et al., 2023). This powder is subsequently refined to craft PG-based construction materials (Rashad, 2017). Currently, the most

substantial application of PG recycling in China is construction materials, accounting for 15.52% of PG recycling in the surveyed region. PG-based construction materials mainly include plastering mortar, self-leveling mortar, blocks, and faced plasterboard, etc. (Yang et al., 2016; Cao et al., 2022). The raw material quality and purification treatment degree of PG determine the quality of construction materials (Calderón-Morales et al., 2021). To enhance the quality of PG-based construction materials, relevant companies have been actively pursuing technological and product innovations. Company J in Guizhou Province has established an unfaced plasterboard production line with an annual capacity of 30 million square meters. This facility has the capacity to utilize 50.4×10^4 t of PG annually. Unfaced plasterboard is an innovative construction material that uses high-temperature gypsum as its primary component and various fibers as reinforcement. Compared with faced plasterboard, unfaced plasterboard eliminates the need for protective cardboard, effectively addressing issues like easy peeling, fire resistance, and sound insulation associated with the faced plasterboard. It possesses traits of being flame-retardant, moisture-resistant, and exceptionally strong, making it well-suited for use in partition walls, ceilings, and decorative applications in various environments such as hotels, schools, hospitals, office buildings, and more.

(3) Engineering application

The engineering applications of PG mainly include waterproof damming materials and road paving materials (Jin et al., 2023). The promotion of engineering application is centered in Guizhou and Yunnan Provinces. Company J in Guizhou Province utilizes 200×10^4 t of PG for waterproof damming materials. In Yunnan Province, both Company O and Company R utilize 49.3×10^4 t and 113.11×10^4 t of PG respectively for waterproof damming materials. Engineering application can realize the one-time consumption of a large amount of PG (Pu

et al., 2021). Beyond waterproof damming materials, Company H in Hubei Province, along with Company N and Company Q in Yunnan Province is conducting demonstration trials involving PG as road paving materials, aiming to achieve broader engineering implementation and large-scale utilization in the future.

(4) Polymer materials

The modification process of PG in polymer materials includes high-temperature calcination to eliminate moisture and impurities, resulting in inert pyrophosphate formation. This pyrophosphate is further subjected to ball milling to produce PG powder in the range of 800–1200 mesh. An additional 5–10% of auxiliary materials are incorporated into the PG powder to adjust its PH to a neutral level, ensuring complete compatibility with polymer materials (Zhang et al., 2022). This innovation offers the potential to replace calcium carbonate powder in polymer materials. Company K in Guizhou Province is dedicated to researching and advancing the utilization of PG in polymer materials. Company M in Yunnan Province has partnered with Company K to collectively promote the adoption of PG in polymer materials, resulting in the successful establishment of a pilot program. This utilization not only enhances the durability and strength of polymer materials, but also diminishes the requirement for extracting calcium carbonate powder, consequently mitigating dust pollution and elevating the resource utilization efficiency of PG (Wu et al., 2022a). Its offerings find primary application in municipal infrastructure construction, encompassing an array of products such as manhole covers, water grates, power pipes, tree guards, isolation fences, water meter boxes, sockets, PE drainage pipes, communication pipes, and more. Given the elevated value of polymer materials, their sales are scarcely constrained by transportation distances, resulting in a wide-ranging and promising market outlook.

(5) Underground filling and ecological restoration

Underground filling and ecological restoration can rapidly and extensively achieve PG recycling. Company J can utilize 300×10^4 t of PG through underground filling, which accounts for approximately 26.09% of the yearly PG production within the surveyed region of Guizhou Province, as shown in Fig. 2. The main ingredient, yellow phosphorus slag, is combined with other ingredients like PG and siliceous materials (Liu et al., 2024). After measuring and grinding, these ingredients are transformed into filling materials suitable for underground filling. This application offers benefits such as extending the operational lifespan of phosphate mines, averting geological hazards, and addressing the environmental contamination associated with stockpiling PG. Compared with underground filling, ecological restoration places a greater emphasis on rejuvenating ecosystems that have been compromised in polluted areas (Chen et al., 2022). Company M and Company N in Yunnan Province are focused on the research and development (R&D) of PG. After harmless treatment, PG is combined with an activator and a cementing agent to form a PG-based ecological restoration material.

(6) Others (Sulfur acid, fertilizers, soil amendment, crafts)

Company I in Guizhou Province harnesses the calcium sulfate present in PG to generate sulfuric acid (Ma et al., 2022). This process involves utilizing an annual quantity of 89.96×10^4 t of PG, representing 7.82% of the yearly PG production within the surveyed region of Guizhou Province, as shown in Fig. 2. Furthermore, PG finds application in the agricultural domain (Papastefanou et al., 2006). Both Company I and Company J in Guizhou Province allocate a portion of PG for fertilizer production, with annual consumption amounts of 11×10^4 t and 100×10^4 t, respectively. Company N in Yunnan Province has successfully tested PG as a soil amendment, confirming that it can enhance soil structure, augment soil fertility, and stimulate crop growth. Additionally, Company M and Company Q in Yunnan Province are currently exploring the application of PG for making arts and crafts, which is expected to become an innovative path for PG recycling.

4.2. Drivers on the development of phosphogypsum recycling industry

Through field research in surveyed regions, this study, based on the actual circumstances of PG recycling industry in China, has summarized the drivers influencing the development of PG recycling industry. These drivers include types of companies, technological innovation, products and markets, industrial policies, geographical location, as shown in Fig. 3.

(1) Types of companies

This study encompassed 18 surveyed companies, categorized into PG production companies, PG production & recycling companies, and PG recycling companies. PG production companies refer to phosphorus chemical companies whose industrial by-product of producing phosphorus compound fertilizers is PG (Mahmoud et al., 2023). Aiming to enhance PG recycling, related companies have also emerged, jointly promoting the development of PG recycling industry.

- Integrated PG production & recycling companies. Confronted with the imperative of compulsory PG consumption, the majority of phosphorus chemical companies must endeavor to harness PG as a resource. This drive leads to the establishment of companies that integrate both PG production and recycling. When such a company is small-scale, like Company A in Sichuan Province, it facilitates the swift utilization of PG. However, due to cost and technical constraints, the applications of such companies are mainly conventional cement retarders and construction materials. When such a company is relatively large-scale, like Company J in Guizhou Province, the significant production of PG prompts the company to enhance its technological R&D endeavors. This strategic approach contributes to the exploration of novel applications.
- Independent PG recycling companies. Amid the development of PG recycling industry, certain entrepreneurs have established autonomous companies dedicated to PG recycling, driven by environmental concerns or economic incentives. These companies collaborate with PG production companies, bolstered by upstream support, thus establishing a comprehensive supply chain. Such companies predominantly concentrate on technological R&D, and innovation within the realm of PG recycling. For example, Company H in Hubei Province is dedicated to enhancing the technological aspects of employing PG in construction materials, while Company K in Guizhou Province specializes in applying PG to polymer materials.
- Inter-regional cooperative companies for PG recycling. Collaboration between Company M in Yunnan Province and Company K in Guizhou Province led to the formation of a joint venture dedicated to inter-regional PG recycling. Leveraging the advanced technology of Company K, the partnership facilitated the collaborative development and production of environmentally friendly municipal products such as, roadside stones, water grates, and manhole covers. This initiative contributes to addressing Yunnan Province's challenge of stagnant PG recycling.

(2) Technological innovation

Fostering technological innovation stands as the principal catalyst propelling the advancement of the high-quality evolution within PG recycling industry (Zhang et al., 2023). This technological innovation primarily encompasses two phases of PG production and PG recycling. Technological breakthroughs in different phases are of great significance to the quality of PG and PG-based products.

- The semi-hydrate wet process phosphoric acid technology during the PG production phase. Company C in Sichuan Province adopted the semi-hydrate wet process phosphoric acid technology. This implementation enables phosphorus chemical companies to achieve phosphoric acid concentrations exceeding 42%, along with a

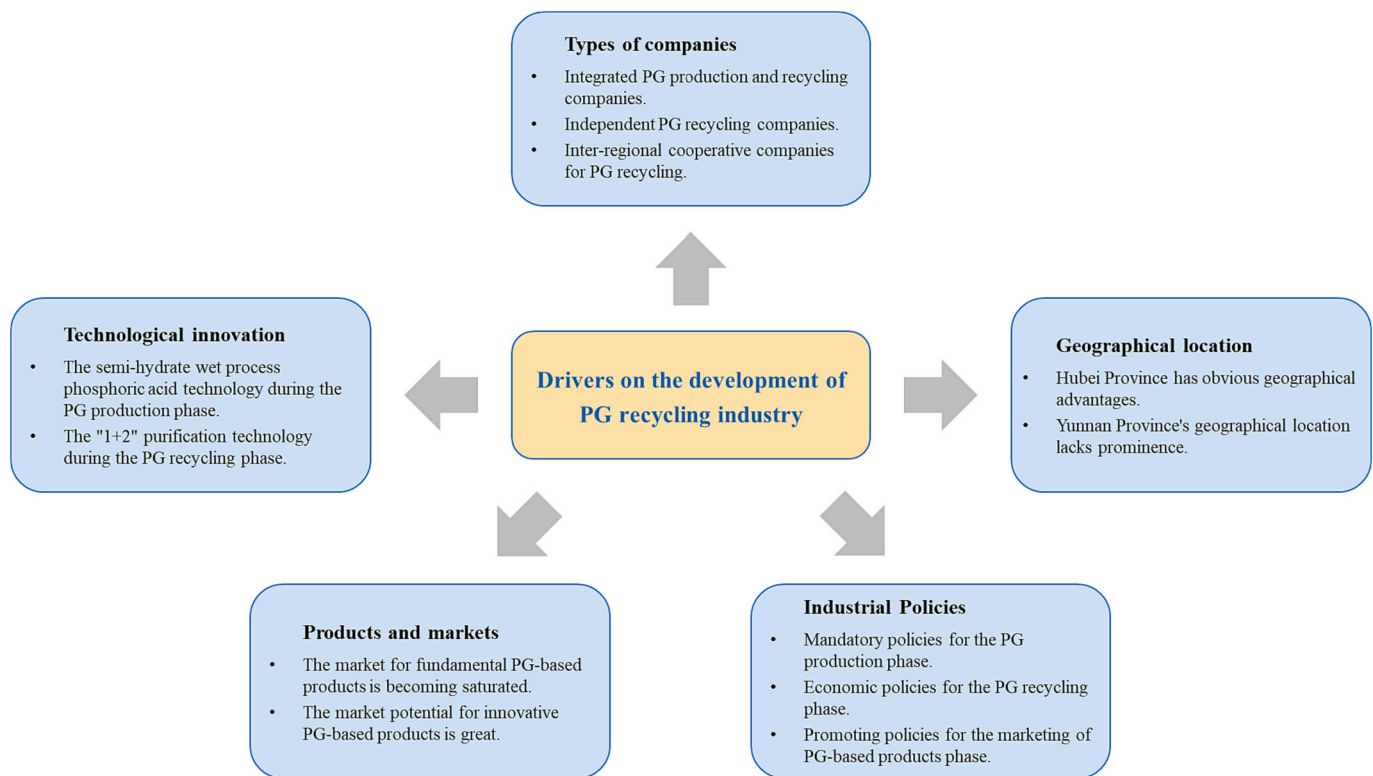


Fig. 3. Drivers on the development of PG recycling industry.

phosphorus yield surpassing 95.2%. This process not only yields premium-quality phosphoric acid but also reduces production costs by 15% to 20% in comparison to the dihydrate wet process phosphoric acid technology. The by-product generated by phosphorus chemical companies are transformed from dihydrate PG to α -hemihydrate PG. α -hemihydrate PG exhibits attributes such as elevated strength, extended setting duration, and economical processing (Du et al., 2022), making it suitable for crafting high-profit self-leveling mortar and plastering mortar.

The "1 + 2" purification technology during the PG recycling phase. The existence of toxic and hazards in PG is one of the main reasons for limiting its recycling. Thus, harmless treatment methods are crucial. Company Q in Yunnan Province adopted the "1 + 2" purification technology. Following one initial wash and two subsequent refined washes, the PG underwent harmless calcination, yielding a product with 75% whiteness in the form of β hemihydrate white gypsum powder. This product adheres to the criteria necessary for crafting premium-quality gypsum-based construction materials.

(3) Products and markets

The crucial stage for the development of PG recycling industry is the marketization of PG-based products. Products and markets are fundamental drivers in influencing the sustainability, economic performance, innovation, and competitiveness of industrial development (Besklubova et al., 2023).

The market for fundamental PG-based products is becoming saturated. Based on the surveyed region, the primary PG applications in China is still fundamental PG-based products such as cement retarders and construction materials. However, in certain surveyed regions like Sichuan, Hubei Province, the market for fundamental PG-based products has been basically saturated, influenced by factors like regional disparities, resource variations, and economic development levels. Furthermore, the Chinese real estate market has

experienced sluggishness in recent years (Su and Qian, 2022), resulting in limited potential for subsequent market expansion.

The market potential for innovative PG-based products is great. For example, Company K in Guizhou Province is actively advocating the utilization of PG in polymer materials. This innovative application allows for the production of municipal infrastructure items, including manhole covers, water grates, power pipes, and drainage pipes, etc. Such products offer substantial value, significant profit potential, and virtually limitless transportation reach. Another example is that Company M and Company Q in Yunnan Province process modified PG instead of natural gypsum into arts and crafts. This choice has garnered substantial market demand and holds considerable market potential.

(4) Industrial policies

Due to its prominent environmental conservation characteristics, the establishment, development and transformation of PG recycling industry are inseparable from policy guidance (Cui et al., 2022b). Policies have wielded an indispensable influence across every phase of PG production, PG recycling, and the marketing of PG-based products.

Mandatory policies for the PG production phase. In 2018, Guizhou Province took the pioneering step of introducing the "determining production by slag" policy, directly connecting the PG production from phosphorus chemical companies with the production of phosphoric acid and phosphate fertilizers (Guizhou Provincial People's Government, 2018). Guided by the extended producer responsibility system, phosphorus chemical companies are compelled to strengthen the PG recycling. This involves achieving a balance between PG production and consumption, thereby effectively mitigating the continued expansion of PG stockpiles.

Economic policies for the PG recycling phase. Different provinces and cities have established varied economic policy standards for PG recycling. Economic policies primarily encompass special subsidies for PG recycling companies, subsidies and tax incentives for PG-

based products, etc. For example, the Ministry of Finance has stipulated that PG-based products can be refunded by 70% from 2019 (Ministry of Finance of the People's Republic of China, 2019). Another example is Deyang City in Sichuan Province, which has introduced an incentive plan. Under this plan, PG recycling companies will receive a subsidy of 1 RMB per ton based on their actual annual utilization. Furthermore, companies engaged in the construction or renovation of production lines will be eligible for special subsidies of up to 2 million RMB. Additionally, PG recycling companies will be given priority access to preferential policies, including securing production factors such as direct electricity purchase and gas supply (Deyang Municipal People's Government, 2017).

Promoting policies for the marketing of PG-based products phase. Given the limited market recognition and sales challenges associated with PG-based products, there is an immediate need for promoting policies to guide the market. In 2018, Guizhou Province put forward an initiative to actively encourage the utilization of PG-based products that meet quality standards and usage criteria in various government projects. These projects encompass municipal engineering construction, transportation development, government-subsidized

housing construction, and village reconstruction (Guizhou Provincial People's Government, 2018).

(5) Geographical location

Geographical location plays a crucial role in shaping the developmental trajectory and potential of a region's PG recycling industry, serving as a fundamental requirement for regional advancement (Wu et al., 2022b). The significant variations in geography between Hubei Province and Yunnan Province are clearly illustrated in Fig. 1.

- Hubei Province has obvious geographical advantages. The PG recycling industry in Hubei Province is mainly concentrated in Yichang City, strategically positioned at the confluence of the upper and middle sections of the Yangtze River. The city boasts an intricate water network, exceptional port facilities, and effortless waterway connectivity, presenting distinct geographical strengths. Leveraging these advantages, the PG recycling industry can capitalize on cost-effective transportation, rapidly accessing markets in cities along the Yangtze River.
- Yunnan Province's geographical location lacks prominence. The PG recycling industry in Yunnan Province is mainly concentrated in

Drivers		The PG production phase	The PG recycling phase	The marketing of PG-based products phase
Types of companies	Obstacles	• Phosphorus chemical companies lack recognition of PG recycling.	• A shortage of innovative PG recycling companies.	—
	Solutions	• Seeking the transformation and upgrading of phosphorus chemical companies. • Considering PG as a resource for reuse rather than industrial solid waste.	• Promoting the coordinated development of PG recycling companies. • Accelerating research and development of premium PG-based products.	—
Technological innovation	Obstacles	• An absence of well-defined evaluation standards for PG recycling technologies.	• An absence of well-defined evaluation standards for PG recycling technologies. • A deficiency in innovative high-value technologies.	—
	Solutions	• Establishing a standardized system for evaluating PG recycling technologies.	• Establishing a standardized system for evaluating PG recycling technologies. • Enhancing the innovation and dissemination of high-value technologies.	—
Products and markets	Obstacles	—	—	• PG-based products face limited market recognition and pose challenges in terms of marketing.
	Solutions	—	—	• Broadening market channels for PG-based products. • Forming a reproducible and scalable PG recycling industrial development model.
Industrial policies	Obstacles	• A lack of coordination among various government departments at the provincial, city, and county levels. • The designated target for PG recycling rate is excessively ambitious.	• A lack of specific guidelines regarding how to allocate, execute, and manage dedicated funds. • The current tax preferential policies do not provide adequate support to the PG-based products.	• The current supporting policies and measures do not provide sufficient incentives for consumers to adopt PG-based products.
	Solutions	• Reinforcing the top-level policy design. • Enhancing the extended producer responsibility system.	• Monitoring the implementation of the industrial policies rigorously. • Updating the tax preferential list for PG-based products promptly.	• Offering discounted freight prices to consumers. • Enforcing a minimum utilization rate requirement for PG-based products in related industries.
Geographical location	Obstacles	• The lack of geographical advantages in certain region such as Yunnan Province.	• The lack of geographical advantages in certain region such as Yunnan Province.	• The high transportation costs and restricted distances constrain the promotion of the PG-based products.
	Solutions	• Developing plans suitable for the development of regional PG recycling industry based on local conditions.	• Developing plans suitable for the development of regional PG recycling industry based on local conditions.	• Considering the mid- to long-term growth prospects and avoid hasty attempts to significantly boost the PG recycling rate in the short term.

Fig. 4. Obstacles and solutions on the development of PG recycling industry.

Kunming City. Situated amidst the intricate topography and landscapes of the Yunnan-Guizhou Plateau, Kunming City grapples with intricate terrain. Transportation relies heavily on road networks, incurring elevated transit expenses. Consequently, the majority of PG-based products find their market primarily within the vicinity of surrounding cities.

4.3. Obstacles on the development of phosphogypsum recycling industry

According to the analysis of the drivers on the development of PG recycling industry, despite some advantages have emerged in its progression, challenges persist in the phases of PG production, PG recycling, and the marketing of PG-based products. Various obstacles impede the industry's path toward sustainable and robust growth, as shown in Fig. 4.

(1) Insufficient recognition of PG recycling and a shortage of innovative PG recycling companies.

Over an extended duration, the primary endeavors of Chinese phosphorus chemical companies revolved around phosphoric acid and phosphate fertilizers. The PG, as the by-product of industrial solid waste, its disposal has traditionally involved accumulation through stockpiling methods. Until December 2018, the *Action Plan for the Protection and Restoration of the Yangtze River* was released (Ministry of Ecology and Environment of the People's Republic of China, 2018). Faced with scrutiny from central environmental inspectors, phosphorus chemical companies were compelled to address the matter of PG recycling. The majority of phosphorus chemical companies need to solve this problem by themselves, leading to the emergence of integrated PG production & recycling companies. However, these companies tend to view PG recycling primarily as an environmental responsibility rather than recognizing it as a genuine industry. This limited perspective on PG recycling impedes its industry-wide development. In addition, although independent PG recycling companies and inter-regional cooperative companies for PG recycling have emerged, a significant portion of companies continue to rely primarily on basic applications such as cement retarders and construction materials. Currently, a scarcity of inventive companies specializing in the production of substantial quantities of premium PG-based products exists.

(2) An absence of well-defined evaluation standards for PG recycling technologies and a deficiency in innovative high-value technologies.

With the development of PG recycling industry, numerous associated technologies have emerged during PG production and PG recycling phases, including PG modification, non-hazardous treatment, and PG-based products processing technologies, further forming different applications of recycled PG. However, the existing evaluation standards of PG recycling technologies diverge, and the absence of relevant technical specifications has led to varying quality levels in both PG and PG-based products. Additionally, the current application of PG recycling technologies in China is still mainly to produce low-value products, such as cement retarders and fundamental construction materials like PG blocks. Consequently, there is a serious lack of crucial technologies for large-scale and high-value utilization. From one perspective, the investment cost for technological R&D is exorbitant, leading to limited R&D capabilities within the majority of PG recycling companies. These companies face challenges due to their inadequate access to financial support, which further dampens their motivation for R&D initiatives. Simultaneously, a significant portion of innovative technologies remain in the developmental stage within laboratories. This is primarily due to a lack of widespread market recognition, rendering the task of market promotion and industrialization for these technologies is difficult.

(3) PG-based products face limited market recognition and pose challenges in terms of marketing.

PG contains trace amounts of heavy metals and radioactive elements, the majority of which exist as insoluble solids (Vásconez-Maza et al., 2022). Despite undergoing modifications and harmless treatments, its hazards are negligible. However, PG-based products still struggle to earn the trust and recognition of the public. Currently, the market entry for

most PG-based products depends on government cooperative and demonstration projects. This results in a PG recycling industry chain that lacks the capacity for self-regulation in the market. Because the overall market acceptance of PG-based products is limited, promoting these products in industrial, agricultural, and environmental restoration sectors has proven to be quite challenging. Finding effective strategies to expand the sales of PG-based products has become a formidable challenge for PG recycling companies.

(4) The industrial policies of PG recycling industry need to be improved.

To promote the growth of PG recycling industry, both national and local governments have introduced a series of relevant industrial policies in recent years. While these policies have achieved initial outcomes in terms of PG recycling, there are still numerous shortcomings within the existing policies. During the PG production phase, while the nation has established directional macro policies aimed at the industrial development and transformation of phosphorus chemical companies, there's often a lack of coordination among various government departments at the provincial, city, and county levels, leading to a deficiency in specific supporting policy measures. Additionally, it holds great significance for the government to exercise control and supervise the PG production within phosphorus chemical companies. It is imperative to avoid hastily establish ambitious targets for the PG recycling rate, but rather, to contemplate the long-term development of PG recycling industry. During the PG recycling phase, firstly, while supporting policies have established dedicated funds for PG recycling projects, there is a lack of specific guidelines regarding how to allocate, execute, and manage these funds. This deficiency significantly diminishes the effectiveness of financial support. Secondly, the value of PG-based products is generally modest, and the current tax preferential policies do not provide adequate support. Moreover, numerous PG-based products remain excluded from the list of tax preferential items. During the marketing of PG-based products phase, marketing these products can be challenging, and the current supporting policies and measures do not provide sufficient incentives for consumers to adopt PG-based products.

(5) The lack of geographical advantages in certain regions can hinder the promotion of traditional PG-based products.

Compared with other provinces, the geographical location of Yunnan Province is notably less dominant. PG production in Yunnan Province is mainly concentrated in Kunming City, where the production volume of PG is large. Given Yunnan Province's prevalence of plateau landforms characterized by intricate terrain and relatively sparse population density, the inherent demand for cement and construction materials is not substantial. Nonetheless, China's current applications to utilizing PG primarily revolves around low-value applications like cement retarders and fundamental construction materials. The high transportation costs and restricted distances for transportation significantly constrain the promotion of these products. In regions such as Yunnan Province, substantial geographical limitations pose challenges in sustaining and advancing the market for PG-based products, which in turn makes it difficult to increase the PG recycling rate in Yunnan Province. Thus, it becomes imperative to explore new paths of development to address the constraints imposed by geographical constraints.

4.4. Solutions on the development of phosphogypsum recycling industry

To address the obstacles encountered in the development of PG recycling industry in China, this study proposes specific solutions, as shown in Fig. 4.

(1) Seeking the transformation and upgrading of phosphorus chemical companies and promoting the coordinated development of PG recycling companies.

As China's phosphorus chemical industry undergoes profound supply-side structural reform, numerous regions have strictly restricted the mining of phosphate rock. Consequently, the market's availability of

phosphate rock has notably diminished. Nonetheless, the demand within the phosphoric acid and phosphate fertilizers remains substantial, compelling phosphorus chemical companies need to actively pursue transformation and upgrading measures. A comprehensive shift is required in the foundational operational principles of phosphorus chemical companies, wherein PG should be perceived as a resource for reuse, rather than industrial solid waste. Simultaneously, there is a need to intensify efforts in the extensive exploration of PG recycling, enhance the industrial value chain for PG recycling, and achieve an eco-friendly transformation within the phosphorus chemical industry. Furthermore, the composite path of cross-regional collaboration and multi-industry coordination is better suited for facilitating industrial transformation and upgrading (Dai et al., 2022). It is necessary to break the geographical constraints of PG recycling companies, facilitate cooperative growth with industries both upstream and downstream across diverse regions, draw insights from technologically advanced high-quality companies, accelerate R&D of premium PG-based products, address existing PG stock, and promote the rapid expansion of cross-regional interconnected industries.

(2) Establishing a standardized system for evaluating PG recycling technologies and enhancing the innovation and dissemination of high-value technologies.

Given the existing issue of inconsistent evaluation standards for PG recycling technologies, there is a pressing need to swiftly establish a comprehensive evaluation standard system. The evaluation standard system should encompass national, industry, local, and enterprise standards, facilitating harmonization and alignment across various levels of standards and providing guidance for technological innovation. Next, there is a need to intensify the innovation and dissemination of high-value technologies. Initially, it is essential to leverage diverse national or local scientific and technological innovation platforms to facilitate the integration of industry, academia, and research. This involves establishing collaborative technological R&D teams, where companies assume the primary role and universities or research institutes contribute as auxiliary components, fostering joint efforts to address pivotal technological challenges. Secondly, enhancing financial backing is imperative. This involves prioritizing investments in or support for crucial technological R&D. And there should be an augmentation of support for the application and transformation of patents related to high-value technologies. Thirdly, dismantling information barriers through the utilization of the internet. This includes reinforcing the understanding and recognition of crucial technologies among relevant companies, thereby driving the industrialization of high-value technologies.

(3) Broadening the market channels for PG-based products and forming a replicable and scalable PG recycling industrial development model.

Confronted with the challenge of promoting PG-based product markets, it is very important to broaden the market channels for PG-based products. First, through engineering application, the problem of low-grade PG recycling can be solved in large quantities. Concurrently, it can increase the proportion of high-grade PG-based products in the market to prevent inferior products from displacing superior ones. Second, government departments should be encouraged to give priority to PG-based products in procurement and increase the demonstration role of the government. Third, it's essential to leverage internet media extensively to overcome information barriers, intensify the promotion of PG-based products, broaden the product market scope, encourage environmentally friendly consumption, and stimulate circular economy development within PG recycling industry, with a focus on demand-side strategies. Furthermore, it is imperative to promptly establish a replicable and scalable PG recycling industrial development model. The PG recycling industrial development model refers to the general strategy covering the entire industrial chain with the goal of PG recycling. It aligns with the strategic objectives and interests of diverse stakeholders and employs a combination of market regulation and government

intervention to optimize economic benefits throughout the PG production, PG recycling, and the marketing of PG-based products phases. The establishment of PG recycling industrial development model facilitates the identification of appropriate roles for all stakeholders, allowing them to leverage their individual strengths. This fosters stable partnerships, enhances industrial efficiency, and, as a result, boosts the market recognition of PG-based product markets.

(4) Accelerating the enhancement of the policy system for PG recycling industry.

The establishment and enhancement of the policy system for PG recycling industry are crucial assurances for ensuring its sustainable and robust development. During the PG production phase, firstly, it is essential to reinforce the top-level policy design. And it is imperative to fortify national-level legislation, systematically merge local policies and regulations, and establish comprehensive institutional mechanisms to support local policy development. Secondly, the extended producer responsibility system must be enhanced. This approach compels phosphorus chemical companies to prioritize the reduction, reuse, and recycling of PG. During the PG recycling phase, on the one hand, it is essential to monitor the implementation of the industrial policies rigorously. This includes conducting quantitative assessments of policy effectiveness and assigning dedicated personnel to conduct regular visits and investigations. On the other hand, it is crucial to promptly update the tax preferential list for PG-based products. This action aims to incentivize the widespread adoption of PG-based products in municipal engineering construction. During the marketing of PG-based products phase, offering discounted freight prices to consumers can help expand the sales reach. In addition, it is advisable to enforce a minimum utilization rate requirement for PG-based products in related industries, alongside implementing incentives and regulatory measures for the construction and real estate sectors. These actions are intended to boost market demand for PG-based products. The government should utilize policy instruments to guide market mechanisms, ultimately promoting the long-term growth and marketization of PG recycling industry.

(5) Developing plans suitable for the development of regional PG recycling industry based on local conditions.

In provinces such as Yunnan, where PG production is high but the demand for PG-based products is limited, it is essential to formulate a new development approach for the local PG recycling industry, in line with the region's specific circumstances. This strategy should consider the industry's mid- to long-term growth prospects and avoid hasty attempts to significantly boost the PG recycling rate in the short term. In the short term, one potential approach is to utilize PG on a large scale to address the challenge of filling numerous privately excavated and unregulated mine pits in the region, thereby mitigating the scarcity of pit-filling materials. In the medium term, PG can be stockpiled harmlessly to minimize the environmental impacts of PG. Simultaneously, this period allows for an ample time to intensify efforts in exploring high-value technologies. In the long term, the production of acid, chemical materials, or other high-value PG-based products can be considered.

5. Conclusions, challenges, and prospects

Throughout the 14th Five-Year Plan period, the development of PG recycling industry stands as a vital component of comprehensive commitment to sustainable development in China. The sustainable and robust growth of PG recycling industry is poised to enhance the environmental conditions within the Yangtze River, facilitate the green transition of the phosphorus chemical industry, and propel the advancement of ecological civilization in China. By conducting an extensive investigation of 18 prominent companies engaged in PG recycling across the provinces of Sichuan, Hubei, Guizhou, and Yunnan, this study comprehensively analyzes applications, drivers behind the progression of PG recycling industry in China. It also summarizes the obstacles and corresponding solutions encountered at various phases of PG production, PG recycling, and the marketing of PG-based products.

The ultimate aim is to offer valuable insights and benchmarks for the advancement of PG recycling industry not only within China but on a global scale.

This study serves as a foundational exploration of applications, drivers, obstacles, and solutions within the development of PG recycling industry. Future research should focus on the quantitative evaluation of the advancement of PG recycling industry. For example, by conducting a quantitative evaluation of the comprehensive benefits from the current development of PG recycling industry, including economic benefits, social benefits and environmental benefits, we can subsequently formulate enhancement strategies grounded in the evaluation results. In addition, it is imperative to consider the influence of policies on the progression of PG recycling industry. We should explore the implementation effects of related policies, offering a theoretical foundation for policymaking by government departments at all levels.

CRedit authorship contribution statement

Yue Cui: Formal analysis, Writing – original draft. **Jiandong Bai:** Visualization, Investigation, Data curation. **I-Shin Chang:** Conceptualization, Validation, Writing – review & editing. **Jing Wu:** Writing –

review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

Appendix Table A

Survey data of PG recycling industry in China.

No.	Companies	Types of companies	Location	Production (10,000 t/y)	Recycling (10,000 t/y)	Applications	Products	Recycling (10,000 t/y)	Sales range (km)	Other information (PG properties or harmless treatment methods)
1	Company A	PG production & recycling	Mianyang, Sichuan	9.2	67	Cement retarder Construction materials	Cement retarder Gypsum powder	33.5 33.5	100 100	Harmless treatment method of modifier neutralization.
2	Company B	PG production & recycling	Deyang, Sichuan	42.2	44.6	Cement retarder Construction materials	Cement retarder Gypsum powder, gypsum blocks, plastering mortar, faced plasterboard, etc.	22.3 22.3	500 300	Harmless treatment method of modifier neutralization.
3	Company C	PG production & recycling	Deyang, Sichuan	20	20	Cement retarder Construction materials	Cement retarder Plastering mortar, self-leveling mortar, etc.	18 2	100 300	PG modified into semi-hydrated PG is applied in construction materials. Harmless treatment method of washing and modifier neutralization.
4	Company D	PG recycling	Chengdu, Sichuan	–	46.18	Construction materials	Gypsum powder, faced plasterboard, etc.	46.18	600	Harmless treatment method of modifier neutralization.
5	Company E	PG production	Yichang, Hubei	150	–	–	–	–	–	–
6	Company F	PG production & recycling	Yichang, Hubei	150	80	Cement retarder Construction materials	Cement retarder Gypsum powder	60 20	1000 1000	Harmless treatment method of modifier neutralization.
7	Company G	PG production & recycling	Yichang, Hubei	160	64	Cement retarder	Cement retarder	64	700	Harmless treatment method of washing.
8	Company H	PG recycling	Yichang, Hubei	–	60	Construction materials Engineering application	Gypsum powder, gypsum blocks, plastering mortar, self-leveling mortar, etc. Road paving materials	60 (P ⁺)	1100 –	Harmless treatment method of washing and modifier neutralization.
9	Company I	PG production & recycling	Qiannan, Guizhou	150	150	Cement retarder	Cement retarder	15.29	150	Harmless treatment method of washing and high-

(continued on next page)

Appendix Table A (continued)

No.	Companies	Types of companies	Location	Production (10,000 t/y)	Recycling (10,000 t/y)	Applications	Products	Recycling (10,000 t/y)	Sales range (km)	Other information (PG properties or harmless treatment methods)
10	Company J	PG production & recycling	Guiyang, Guizhou	1000	900	Construction materials	Gypsum powder, gypsum blocks	33.75	150	temperature calcination.
						Fertilizers	Fertilizer of calcium silicon magnesium potassium	11	–	
						Sulfur acid	Sulfur acid	89.96	–	
						Cement retarder	Cement retarder	150	150	
						Construction materials	Gypsum powder, plastering mortar, gypsum blocks, plaster cast, unfaced plasterboard, etc.	150	1800	Harmless treatment method of washing and high-temperature calcination.
						Engineering application	Waterproof damming materials	200	–	
						Underground filling	Filling materials	300	–	
11	Company K	PG recycling	Guiyang, Guizhou	–	31	Polymer materials	Fertilizer of calcium silicon magnesium potassium	100	–	Modified anhydrous PG. Harmless treatment method of high-temperature calcination.
							Polyvinyl chloride power pipes and W-type drainage pipe, polyethylene W-type drainage pipe, composite resin manhole covers, and PG curb stone	31	1400	
12	Company L	PG production	Kunming, Yunnan	1300	–	–	–	–	–	–
13	Company M	PG recycling	Kunming, Yunnan	–	48.64	Cement retarder	Cement retarder	27	300	PG modified into semi-hydrated PG is applied in construction materials and anhydrous PG is applied in polymer materials. Harmless treatment method of washing (a combination of warm water washing and lime water washing) and high-temperature calcination.
						Construction materials	Gypsum powder, gypsum blocks, plastering mortar, self-leveling mortar, thermal insulation mortar, etc.	21.64	600	
						Polymer materials	Unsaturated polyester resin, polyethylene, polypropylene, polyvinyl chloride, rubber agricultural mulch film	(P ^a)	–	
							Base material for filling	(P ^a)	–	
						Ecological restoration	Gypsum crafts	(P ^a)	–	
						Crafts	Cement retarder	55	300	
						Cement retarder	Cement retarder	55	300	
14	Company N	PG production & recycling	Kunming, Yunnan	336.55	96.97	Construction materials	Gypsum powder, gypsum blocks, plastering mortar, faced plasterboard, etc.	41.97	500	Harmless treatment method of washing and calcination.
						Engineering application	Road paving materials	(P ^a)	–	
						Ecological restoration	Base material for filling	(P ^a)	–	
						Soil amendment	Soil amendment	(P ^a)	–	PG modified into semi-hydrated PG is applied in construction materials. Harmless treatment method of washing and low-temperature calcination.
						Construction materials	α semi-hydrated gypsum powder	4.88	500	
15	Company O	PG production & recycling	Kunming, Yunnan	155.62	54.18	Engineering application	Waterproof damming materials	49.3	–	Harmless treatment by upstream companies.
16	Company P	PG recycling	Kunming, Yunnan	–	18	Construction materials	Faced plasterboard	18	500	

(continued on next page)

Appendix Table A (continued)

No.	Companies	Types of companies	Location	Production (10,000 t/y)	Recycling (10,000 t/y)	Applications	Products	Recycling (10,000 t/y)	Sales range (km)	Other information (PG properties or harmless treatment methods)
17	Company Q	PG recycling	Kunming, Yunnan	–	67	Construction materials	β hemihydrate white gypsum powder, plastering mortar, gypsum blocks, etc.	67	1000	PG modified into semi-hydrated PG is applied in construction materials. Harmless treatment method of washing (the “1 + 2” purification technology) and low-temperature calcination.
						Engineering application	Road paving materials	(P ^a)	–	
						Crafts	Gypsum crafts	(P ^a)	–	
18	Company R	PG recycling	Kunming, Yunnan	–	178.27	Cement retarder	Cement retarder	47.23	300	Harmless treatment method of washing and calcination.
						Construction materials	α semi-hydrated gypsum powder, gypsum blocks, plastering mortar, etc.	17.93	–	
						Engineering application	Waterproof damming materials	113.11	–	

^a P represents pilot.

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